Stewart and Greenhalgh: Shadow Prices and Agricultural Land

1 Motivation: What is an additional acre worth?

Reading: Fred J. Stewart and Richard A. Greenhalgh (1977), *The Use of Shadow Prices in Determining Marginal Values for Agricultural Land*. The numerical model below uses Table 1 of the paper; the resource-package argument appears on printed pages 3–7 and in footnote 1.

Land development projects, including drainage, flood protection, and erosion control, can expand the agricultural resource base. A planner needs to compare the benefits of those additions with their development costs. Market evidence can be limited, and a land price need not reveal the contribution of another acre to a particular production system.

Stewart and Greenhalgh ask how an optimization model can estimate that marginal contribution. Their example holds agricultural production requirements fixed and minimizes the cost of meeting them. Additional useful land creates value when it allows the same output to be produced at lower cost.

The complication is that **land is a package of characteristics**. An acre can bring a total-acreage allowance, limits on suitable crop uses, and a requirement to maintain some pasture. Expanding the resource changes several constraints at once. The shadow price on the total-land constraint values only one of those changes.

This application connects duality to an economic question: *Which right-hand sides actually change when a project adds a unit of the resource?* The answer determines which shadow prices belong in its marginal valuation.

The paper develops a small illustrative model, rather than estimating a model from a documented farm dataset. Its values are conditional on the stated production requirements, technologies, costs, and land-use restrictions. They measure production-cost savings over the model's accounting period, not a capitalized market sale price for land.

2 The agricultural production model

2.1 Activities and costs

There are three land groups, indexed by $g = 1, 2, 3$, and three activities on each group. Let C_g, W_g, and P_g denote acres of corn, wheat, and pasture, respectively. These nine acreages are the baseline decision variables.

The full numerical objective is

$$\begin{aligned}\min_{C_1,W_1,P_1,C_2,W_2,P_2,C_3,W_3,P_3} \quad & 66.40C_1 + 96.40W_1 + 80.14P_1 \\ & + 151.20C_2 + 93.80W_2 + 84.16P_2 \\ & + 102.60C_3 + 61.50W_3 + 75.48P_3.\end{aligned}$$

Each coefficient is a cost per acre. The objective is total production cost, measured in dollars. The three lines group costs by land type.

2.2 Production requirements

The model must produce the specified quantities of corn, wheat, and pasture output:

$$\begin{aligned} 80C_1 + 60C_2 + 40C_3 &= 7{,}500 \quad &&\text{(corn)}, \\ 40W_1 + 35W_2 + 25W_3 &= 2{,}500 \quad &&\text{(wheat)}, \\ 2.2P_1 + 2.0P_2 + 1.8P_3 &= 170 \quad &&\text{(pasture)}. \end{aligned}$$

The coefficients are output per acre. Table 1 does not explicitly label the physical output units, so we retain its numerical units. These are **equalities**, as in the paper: the model produces the specified output rather than choosing output in response to product prices.

2.3 Land and cropping restrictions

Each land group initially contains 100 acres. Corn and wheat have maximum acreages; pasture has a minimum acreage. All twelve restrictions are

$$\begin{aligned} C_1 + W_1 + P_1 &\le 100 \quad &&\text{(land 1)}, \\ C_1 &\le 60 \quad &&\text{(corn 1 maximum)}, \\ W_1 &\le 20 \quad &&\text{(wheat 1 maximum)}, \\ P_1 &\ge 20 \quad &&\text{(pasture 1 minimum)}, \\ C_2 + W_2 + P_2 &\le 100 \quad &&\text{(land 2)}, \\ C_2 &\le 40 \quad &&\text{(corn 2 maximum)}, \\ W_2 &\le 30 \quad &&\text{(wheat 2 maximum)}, \\ P_2 &\ge 30 \quad &&\text{(pasture 2 minimum)}, \\ C_3 + W_3 + P_3 &\le 100 \quad &&\text{(land 3)}, \\ C_3 &\le 20 \quad &&\text{(corn 3 maximum)}, \\ W_3 &\le 30 \quad &&\text{(wheat 3 maximum)}, \\ P_3 &\ge 30 \quad &&\text{(pasture 3 minimum)}. \end{aligned}$$

Finally, the variable restrictions are

$$C_1, W_1, P_1, C_2, W_2, P_2, C_3, W_3, P_3 \ge 0.$$

Together, the objective, three production equalities, twelve land-use inequalities, and nine nonnegativity restrictions define the full baseline LP. The cropping bounds can represent physical suitability or imposed land-use rules. Their economic justification matters because they help determine the calculated value of development.

2.4 Baseline allocation

Solving this model in R gives the following acreages and a minimized cost of **$24,012.42**. The results and figure in these notes are computed when the document renders.

Land group	Corn	Wheat	Pasture	Total used
1	60.0	17.5	22.50	100.00
2	40.0	30.0	30.00	100.00
3	7.5	30.0	33.61	71.11

Land group 2 uses all 100 acres, with corn and wheat at their upper bounds and pasture at its lower bound. The economic question is how this whole production system would adjust if more of that land package became available.

3 The main insight: Value the complete resource package

3.1 Specify the change before interpreting the shadow price

Let L denote the endowment of land group 2. Keeping its characteristics in the original proportions replaces its four bounds with

$$\begin{aligned} C_2 + W_2 + P_2 &\leq L, \\ C_2 &\leq 0.4L, \\ W_2 &\leq 0.3L, \\ P_2 &\geq 0.3L. \end{aligned}$$

At the baseline, $L = 100$. An additional acre raises the four right-hand sides by 1, 0.4, 0.3, and 0.3. The fractions apply to the **land endowment**; the solver chooses the actual allocation. Raising the pasture minimum tightens a restriction, so the package brings an additional obligation as well as additional opportunities.

3.2 Derive the weighted sum

Write $C^*(L)$ for minimized cost and λ_i for its local change per unit increase in right-hand side b_i. With the original inequality directions, upper-bound resource prices are nonpositive; minimum-requirement prices are nonnegative.

Within a range where the relevant marginal values remain valid,

$$\Delta C^* = \lambda_L \Delta b_L + \lambda_C \Delta b_C + \lambda_W \Delta b_W + \lambda_P \Delta b_P.$$

Substituting the coordinated changes gives

$$\Delta C^* = (\lambda_L + 0.4\lambda_C + 0.3\lambda_W + 0.3\lambda_P)\, \Delta L.$$

Therefore the marginal benefit of the package, measured as a cost saving, is

$$\boxed{MV(L) = -\frac{dC^*(L)}{dL} = -\left(\lambda_L + 0.4\lambda_C + 0.3\lambda_W + 0.3\lambda_P\right).}$$

This is a **weighted sum**, not a weighted average. The weights are the right-hand-side changes per added acre; there is no division by their sum. Using only λ_L would value more total land while holding every cropping bound fixed, which is a different economic experiment.

3.3 A numerical interpretation

One valid set of multipliers for this numerical model, matching the rounded values in the paper's Table 2, gives the following contributions:

Constraint	Cost shadow price	Change per acre	Cost effect
Total land 2	−1.14917	1.0	−1.14917
Corn maximum	−1.55083	0.4	−0.62033

Constraint	Cost shadow price	Change per acre	Cost effect
Wheat maximum	0	0.3	0
Pasture minimum	+1.44250	0.3	+0.43275
Package			−1.33675

An additional acre saves approximately **$1.34** at the margin. The extra pasture requirement offsets some of the gains from additional land and permitted corn acreage. A direct re-solve at $L = 100.001$ gives a cost saving per added acre of 1.33675, confirming the weighted-shadow-price calculation.

Individual multipliers need not be unique at this solution: the land and cropping constraints bind together. A solver can report a different allocation of shadow values across these rows while giving the same value for the coordinated package. A binding constraint can also have a zero shadow price.

The paper's earlier displayed multipliers on printed page 5 imply about $1.34042, a slightly different value. These notes consistently use Table 1's coefficients, which yield $1.33675 and agree with the later range-analysis tables after accounting for the transfer activity's cost.

4 Representing the package with a transfer activity

Define a new decision variable $t \geq 0$ as additional acres of the land group 2 package. Then $L = 100 + t$, and the four constraints become

$$\begin{aligned} C_2 + W_2 + P_2 - t &\leq 100, \\ C_2 - 0.4t &\leq 40, \\ W_2 - 0.3t &\leq 30, \\ P_2 - 0.3t &\geq 30. \end{aligned}$$

These are the coefficients in the paper's **TRANSFER 2** column. For example, $t = 5$ provides 105 total acres, permits up to 42 corn acres and 31.5 wheat acres, and requires at least 31.5 pasture acres. It adds land to the modeled resource base; it does not move land out of groups 1 or 3.

The other constraints and the nine production-cost coefficients stay as specified above. The paper adds a cost of $0.01t$ to the objective and, in its illustrative bounded experiment, adds the row $t \leq 1$. More generally, a bound $t \leq \bar{t}$ lets the analyst vary how much development is allowed.

If k is development cost per acre, the expanded objective is production cost plus kt. The local change in that objective per unit of transfer is

$$k + \lambda_L + 0.4\lambda_C + 0.3\lambda_W + 0.3\lambda_P = k - MV(L).$$

This links the transfer activity's reduced-cost calculation to the package valuation. Expansion is attractive when its gross cost saving exceeds its development cost. The paper's 0.01 activity cost must be accounted for when recovering gross resource value from its reported net activity value; reduced cost signs also depend on the solver's reporting convention and active bound.

The transfer activity makes a coordinated change into a single explicit choice. Its sensitivity range concerns the whole package. Separate one-at-a-time ranges for land, corn, wheat, or pasture do not directly give the allowable range for that coordinated change.

5 Derived demand for the land package

5.1 Trace marginal value as the endowment changes

To construct the derived-demand schedule, vary L, update all four land group 2 bounds, and re-solve the production model. Keep costs, yields, the other land groups, and output requirements fixed. At each solution calculate

$$MV(L) = -\left(\lambda_L(L) + 0.4\lambda_C(L) + 0.3\lambda_W(L) + 0.3\lambda_P(L)\right).$$

These are valuations of specified endowments. In particular, evaluating $L < 100$ describes a smaller endowment, rather than a negative value of the nonnegative development variable t.

Within each optimal LP regime, cost is linear in L and marginal value is constant. When the limiting constraints or optimal production adjustments change, the slope changes. The result is a stepwise marginal-value schedule. At a corner, use the appropriate one-sided value for expansion or contraction.

The embedded R calculation uses `lpSolve`, audits feasibility and cost at every solution, and checks the shadow-price prediction against small numerical changes in L. It locates the plotted corners by intersecting adjacent linear cost segments.

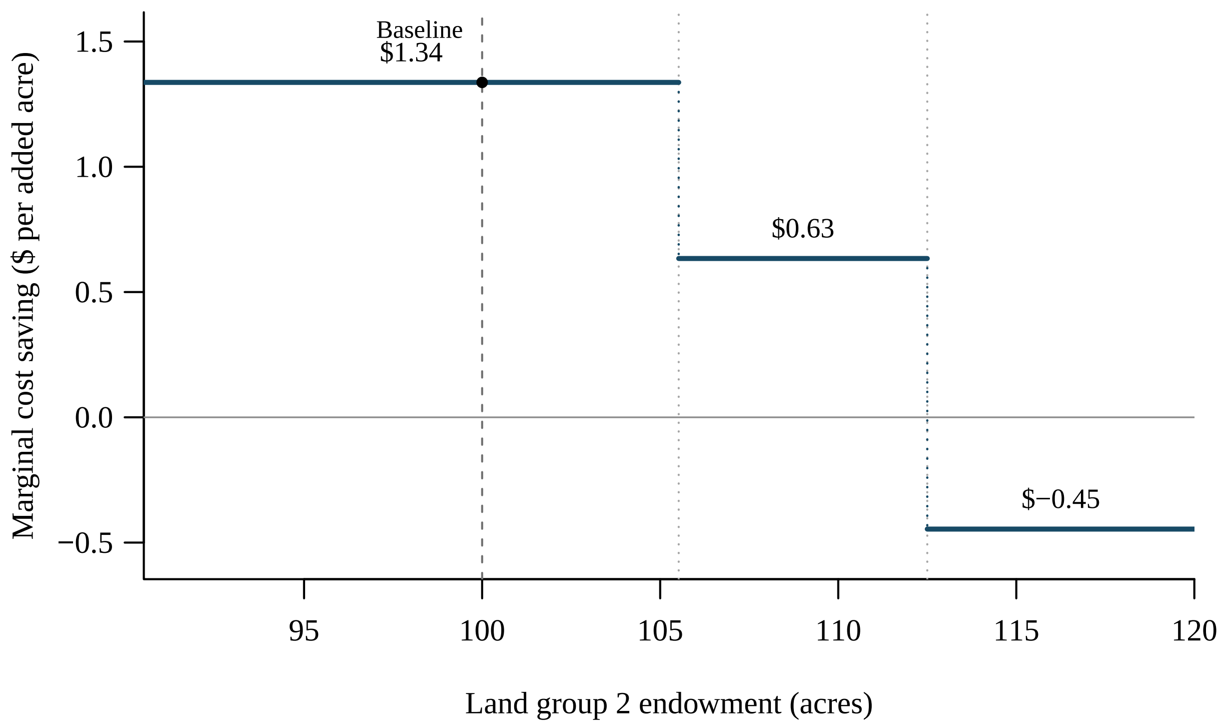

Figure 1: Derived marginal value of the land group 2 package, computed from Table 1. All four land group 2 bounds change together. The dashed line marks the 100-acre baseline; dotted lines mark changes in marginal value. The negative segment reflects the rising pasture minimum.

5.2 Interpret the steps and the development decision

The initial value of approximately \$1.34 persists through about **105.52 acres**. At that point, the minimum pasture acreage on land group 3 binds, changing the production adjustment available to accommodate additional land group 2. The marginal benefit falls to approximately **\$0.63**.

At about **112.50 acres**, corn production on land group 3 reaches zero and the marginal value falls again, becoming negative. Additional land group 2 can no longer save costs by replacing corn on land group 3. The model now pays more to meet the same output requirements as L rises. This is possible because the package increases a compulsory pasture minimum: larger endowments do not simply enlarge the feasible set. The paper's Figure 1 emphasizes the positive steps; the calculation here also displays the next segment implied by its numerical model.

For an optional development project with nonnegative development costs, there is no reason to purchase additions on the negative-value segment. With constant cost k, compare the height of each step with k and expand only while the marginal saving exceeds that cost. For a finite project, total gross benefit is the area under the marginal-value schedule:

$$C^*(100) - C^*(100+t) = \int_{100}^{100+t} MV(L)\, dL.$$

Subtract kt to obtain net benefit. Multiplying the initial shadow value by every added acre is valid only while the project stays within that initial range. The value of the resource follows from both the **package being added** and the **production adjustments available at each endowment**.

6 Model Lab Example

6.1 Question and experimental design

What is the production-cost penalty from applying the land-use restrictions to newly developed group 2 acreage, and how does that penalty change with project size?

Compare two ways of adding t acres of group 2 land. Restricted expansion brings the paper's corn and wheat maxima and pasture minimum. Unrestricted expansion permits any mix of the three activities on the additional land. Both scenarios retain the original restrictions on all existing land, the same yields and costs, and the same output requirements. This isolates the value of flexibility on the new acreage.

Increasing the total-land bound alone while holding the corn and wheat bounds fixed would not implement unrestricted expansion: those crop bounds would still apply to the additional acreage. Instead, distinguish existing and new land explicitly.

Let $z_C, z_W, z_P \geq 0$ denote acres of corn, wheat, and pasture on the **new** group 2 land. The original nine variables describe only existing land. Retain all twelve original land-use inequalities and all nine original nonnegativity restrictions. Replace the output equalities with

$$\begin{aligned} 80C_1 + 60C_2 + 40C_3 + 60z_C &= 7{,}500, \\ 40W_1 + 35W_2 + 25W_3 + 35z_W &= 2{,}500, \\ 2.2P_1 + 2.0P_2 + 1.8P_3 + 2.0z_P &= 170. \end{aligned}$$

If $C_{\text{old}}(x)$ denotes the fully parameterized production-cost objective given above, both scenarios minimize

$$C_{\text{old}}(x) + 151.20z_C + 93.80z_W + 84.16z_P.$$

The **unrestricted** scenario adds only

$$z_C + z_W + z_P \leq t, \qquad z_C, z_W, z_P \geq 0.$$

The **restricted** scenario also imposes

$$z_C \leq 0.4t, \qquad z_W \leq 0.3t, \qquad z_P \geq 0.3t.$$

These are maxima and a minimum, not three required exact shares. The restricted formulation is equivalent, in aggregate, to expanding the original group 2 package to $100 + t$ acres. Because old and new group 2 land have the same technology, labels on otherwise interchangeable acres can be nonunique; the minimized cost is the quantity we use for valuation.

6.2 Define the value of removing the restrictions

Let $C_R(t)$ and $C_U(t)$ be minimized production costs in the restricted and unrestricted scenarios, and let $C_0 = C_R(0) = C_U(0)$ be baseline cost. Gross benefits of development are

$$B_R(t) = C_0 - C_R(t), \qquad B_U(t) = C_0 - C_U(t).$$

The total value of flexibility on the added acreage is

$$\boxed{V(t) = C_R(t) - C_U(t) = B_U(t) - B_R(t).}$$

Every allocation feasible under restricted expansion is also feasible under unrestricted expansion. Therefore $C_U(t) \leq C_R(t)$ and $V(t) \geq 0$ wherever both problems are feasible. The average value per added acre is $V(t)/t$ for $t > 0$; it is distinct from the marginal value $V'(t)$.

No development cost is included in this comparison. An identical cost kt in both scenarios would cancel from $V(t)$. The result is the production system's value of removing the restrictions. Any benefits of the restrictions themselves, such as erosion reduction, would require separate representation.

Define the marginal benefits of acreage as

$$MV_R(t) = -C_R'(t), \qquad MV_U(t) = -C_U'(t).$$

For the new-land constraints, denote cost shadow prices by μ. As in the paper, restricted expansion changes four bounds together, whereas unrestricted expansion changes only the new-land availability bound:

$$\begin{aligned} MV_R(t) &= -\left(\mu_{L,R} + 0.4\mu_{C,R} + 0.3\mu_{W,R} + 0.3\mu_{P,R}\right), \\ MV_U(t) &= -\mu_{L,U}. \end{aligned}$$

Thus, away from corners,

$$\boxed{V'(t) = MV_U(t) - MV_R(t), \qquad V(t) = \int_0^t \left[MV_U(s) - MV_R(s)\right] ds.}$$

Use one-sided slopes at corners and at $t = 0$. In particular, a solver may return a nonunique dual solution when all new-acreage variables are fixed to zero; a small positive expansion identifies the relevant right-hand slope.

6.3 Calculate the value of flexibility

The calculation solves both twelve-variable LPs at each expansion size. It checks feasibility, reconstructs costs, verifies $C_U(t) \leq C_R(t)$, and confirms that restricted expansion reproduces the earlier aggregate model. The table reports dollars over the same accounting period as the original model.

Added acres	Restricted cost	Unrestricted cost	Flexibility value	Value per acre
0	24,012.42	24,012.42	0.00	–
1	24,011.08	24,009.72	1.36	1.36
5	24,005.73	23,998.92	6.82	1.36
10	24,002.20	23,995.04	7.16	0.72
12.5	24,000.61	23,995.04	5.58	0.45
20	24,003.96	23,995.04	8.92	0.45

For a **10-acre project**, restricted expansion saves \$10.22, while unrestricted expansion saves \$17.38. Removing the restrictions is therefore worth **\$7.16 in total**, or **\$0.72 per added acre** on average. This is the maximum additional project cost justified by flexibility alone within this model.

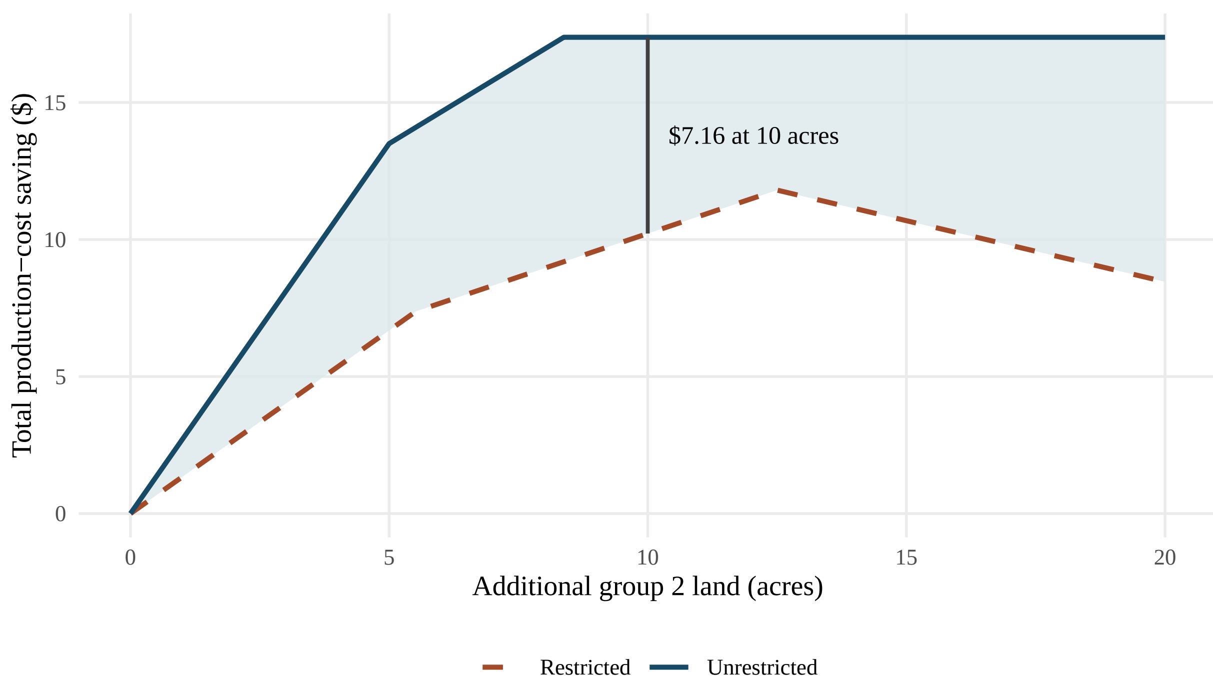

Figure 2: Production-cost savings from adding group 2 land. The vertical gap, shaded between the curves, is the total value of removing restrictions at a given project size. The baseline is the same in both scenarios.

The unrestricted curve levels off after approximately **8.38 added acres**. Beyond that point, more available acreage creates no further production-cost saving. Restricted benefits peak at **12.50 added acres** and then decline as the growing pasture minimum becomes costly. Common development costs would be subtracted from both benefit curves when choosing project size.

6.4 Explain the marginal values and their changing gap

Initially, one unrestricted acre can grow corn and replace $60/40 = 1.5$ acres of corn on group 3. The saving is

$$MV_U(0^+) = 1.5(102.60) - 151.20 = \$2.70 \text{ per added acre.}$$

This adjustment lasts for five added acres, enough to replace the original 7.5 acres of group 3 corn. Restricted expansion initially saves only \$1.33675 per acre, because the corn maximum expands more slowly and the pasture requirement grows. Thus

$$V'(0^+) = 2.70000 - 1.33675 = \$1.36325 \text{ per added acre.}$$

The initial value of flexibility is about \$1.36 per acre, but this rate cannot be multiplied by every acre of a large project.

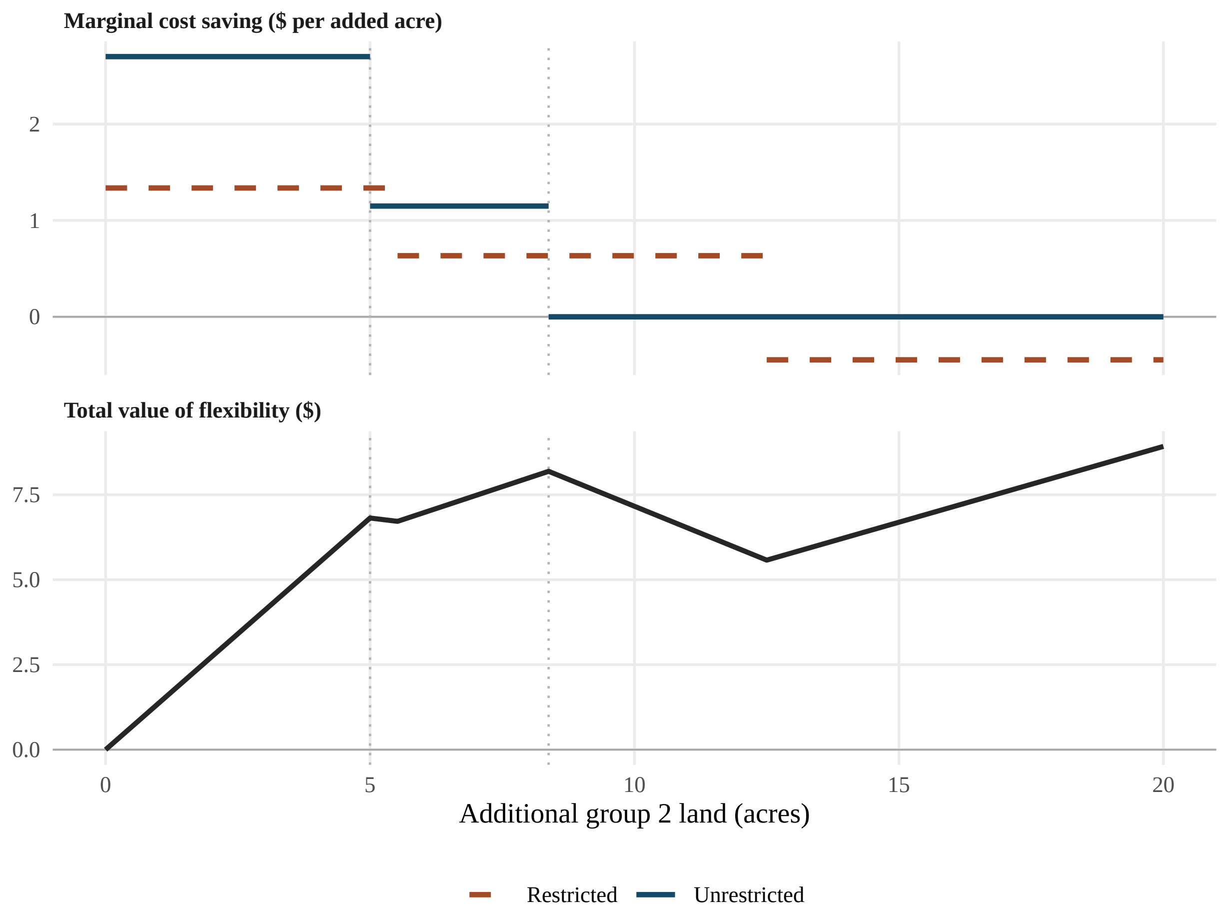

Figure 3: Upper panel: marginal values of added acreage. Lower panel: total value of removing its restrictions. Total flexibility value rises where the unrestricted marginal value is higher and falls where it is lower. Dotted guides mark unrestricted breakpoints; total flexibility value remains nonnegative.

6.5 What the experiment teaches

A nonnegative total value need not increase at every project size. Unrestricted expansion always costs no more than restricted expansion at the same t. Nevertheless, the marginal-value curves cross. Between 5 and about 5.52 added acres, unrestricted corn substitution has ended, while restricted expansion still earns its initial marginal benefit. Between about 8.38 and 12.50 acres, unrestricted benefits are flat while restricted benefits continue rising. In both intervals, $V'(t) < 0$: the restricted scenario partially catches up. That does not imply the restrictions have become beneficial.

After 12.50 acres, the restricted scenario's pasture obligation raises costs, while unrestricted cost stays constant. Consequently the value of removing the restrictions rises again. The integral of the marginal-value gap matches the direct cost difference; the R calculation checks both ways of computing $V(t)$.

Project size and the decision being valued must be explicit. These curves compare two specified projects of equal size. An unrestricted project may leave acreage unused. A restricted project still incurs its pasture minimum on all developed acres, even if it would prefer a smaller project. The plotted comparison stops at $t = 20$: beyond that, the three pasture minima alone imply more than the required 170 units of pasture output, making restricted expansion infeasible under the paper's equality constraint.

For a Model Lab presentation, the central explanation is the sequence from the counterfactual to its consequences: define which land is freed from which rules, predict the initial marginal effect, solve and verify both models, and use changing allocations and binding constraints to explain the difference between the two value curves.